SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – VECTOR DATABASE & SEMANTIC SEARCH BENCHMARK(FAISS VS CHROMADB)

Course Code:	CSA65	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO3 – Precision (BL2, BL3)	**Assessment:**	Assessment Tool 3 – Vector Database & Semantic Search Benchmark(FAISS vs ChromaDB)
Weightage:	10% of CIA	**Total Marks:**	2 * 50 = 100
CO3 Statement:	*Develop intelligent applications using Retrieval-Augmented Generation (RAG), vector databases, and introductory AI agent concepts.(BL3, BL4)*		
Student Name:	Y. Jayesh Ranjan	**Reg. No.:**	192472201
Section / Batch:	CSE – AI	**Date:**	17/08/26

Instructions to Students

- Answer 2 questions. Each question carries 50 marks. Total: 100 marks.
- Implement the solution using Python.
- Use the same dataset, embedding model, and experimental conditions for FAISS and ChromaDB wherever comparison is required.
- Generate embeddings and perform semantic search using FAISS and ChromaDB.
- Demonstrate multiple test queries and record the retrieved results and ranking.
- Compare retrieval relevance, ranking quality, and suitable performance measures such as indexing/search time where applicable.
- Present benchmark results using appropriate tables and/or charts.
- Clearly document the dataset, preprocessing, chunking strategy, embedding model, database configuration, queries, and evaluation methodology.
- Explain the architecture and justify the conclusions drawn from the benchmark.

Code of Conduct

I Y. Jayesh Ranjan – 192472201 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jayesh Ranjan

SECTION A –Vector Database & Semantic Search Benchmark(FAISS vs ChromaDB)

5. Create a vector search system using multiple product manuals. Implement semantic search using FAISS and ChromaDB and evaluate their ability to retrieve relevant sections for installation, troubleshooting, and maintenance queries.
26. Conduct experiments using a small dataset and a large dataset. Compare FAISS and ChromaDB for both datasets and determine whether the preferred database changes with dataset size.

QUESTION 5 – EMPLOYEE POLICY QA SYSTEM

AIM

To develop a Retrieval-Augmented Generation (RAG) based Employee Policy Question Answering system that retrieves relevant information from an employee policy document and answers questions related to leave, working hours, attendance, holidays, and employee benefits.

ALGORITHM

1. Start the program.
2. Create or load the employee policy document.
3. Divide the document into smaller text chunks.
4. Load a sentence-transformer embedding model.
5. Generate embeddings for all document chunks.
6. Store the embeddings in a FAISS vector database.
7. Accept a question from the user.
8. Convert the question into an embedding.
9. Perform semantic search using FAISS.
10. Retrieve the most relevant policy sections.
11. Display the retrieved context and similarity score.
12. Generate an answer using only the retrieved employee policy information.
13. Display the answer along with the source.
14. Repeat the process for multiple queries.
15. Stop when the user enters exit.

CODE

```
# ============================================================
    )



# ============================================================
# 9. MAIN PROGRAM
```

```python
# ============================================================

def main():

    print("\nRAG PIPELINE:")

    print(
        "Employee Policy Document"
        " -> Chunking"
        " -> Embeddings"
        " -> FAISS"
        " -> Retrieval"
        " -> Answer"
    )

    print("\nAsk questions about:")
    print("1. Leave")
    print("2. Working hours")
    print("3. Attendance")
    print("4. Holidays")
    print("5. Employee benefits")

    print("\nType 'exit' to stop.")

    while True:

        query = input(
            "\nEnter your question: "
        )

        if query.lower() == "exit":

            print("\nProgram ended.")
            break
```

```
        if query.strip() == "":

            print(
                "Please enter a question."
            )

            continue

        answer_question(query)


# ============================================================
# START PROGRAM
# ============================================================

if __name__ == "__main__":

    main()
```

INPUT

The following queries can be used to test the system:

How many days of annual leave are employees entitled to?

What are the standard working hours?

What is the attendance policy?

What holidays does the company observe?

What employee benefits are provided?

Enter exit to terminate the program.

OUTPUT

Result:

The program is executed successfully.

QUESTION 26 – RAG WITH IRRELEVANT QUERY DETECTION

AIM

To develop a Retrieval-Augmented Generation (RAG) system that retrieves relevant information from a document and identifies irrelevant or unsupported questions, thereby preventing the system from generating unsupported answers.

ALGORITHM

1. Start the program.
2. Load the knowledge document.
3. Divide the document into smaller chunks.
4. Load a sentence-transformer embedding model.
5. Generate embeddings for all document chunks.
6. Store the embeddings in a FAISS vector database.
7. Accept a query from the user.
8. Generate an embedding for the query.
9. Search the FAISS database for the most relevant chunks.
10. Calculate the similarity distance between the query and retrieved chunks.
11. Compare the best distance with a relevance threshold.
12. If the query is relevant, generate an answer using the retrieved document content.
13. If the query is irrelevant, display that the information is not available in the provided document.
14. Display the retrieved context and source.
15. Repeat the process for multiple queries.
16. Stop when the user enters exit.

CODE

```
# ============================================================
            query,
            results
        )

        print("\n--------------------------------------------")
        print("FINAL ANSWER")
```

```
        print("------------------------------------------")

        print(answer)

        print(
            "\nSource: Operating System Notes"
        )

    else:

        print(
            "STATUS: IRRELEVANT QUERY"
        )

        print("\n------------------------------------------")
        print("FINAL ANSWER")
        print("------------------------------------------")

        print(
            "Information not available in the "
            "provided document."
        )

        print(
            "The system did not generate an "
            "unsupported answer."
        )


# ============================================================
# 10. MAIN PROGRAM
# ============================================================

def main():
```

```
    print("\nRAG PIPELINE:")

    print(
        "Document -> Chunking -> Embedding -> "
        "FAISS -> Retrieval -> Relevance Detection -> Answer"
    )

    print("\nType 'exit' to stop.")

    while True:

        query = input(
            "\nEnter your question: "
        )

        if query.lower() == "exit":

            print("\nProgram ended.")
            break

        if query.strip() == "":

            print(
                "Please enter a question."
            )

            continue

        ask_question(query)


# ============================================================
# START PROGRAM
# ============================================================
```

```python
if __name__ == "__main__":

    main()
```

INPUT

Relevant queries:

What is paging?

What is FIFO page replacement?

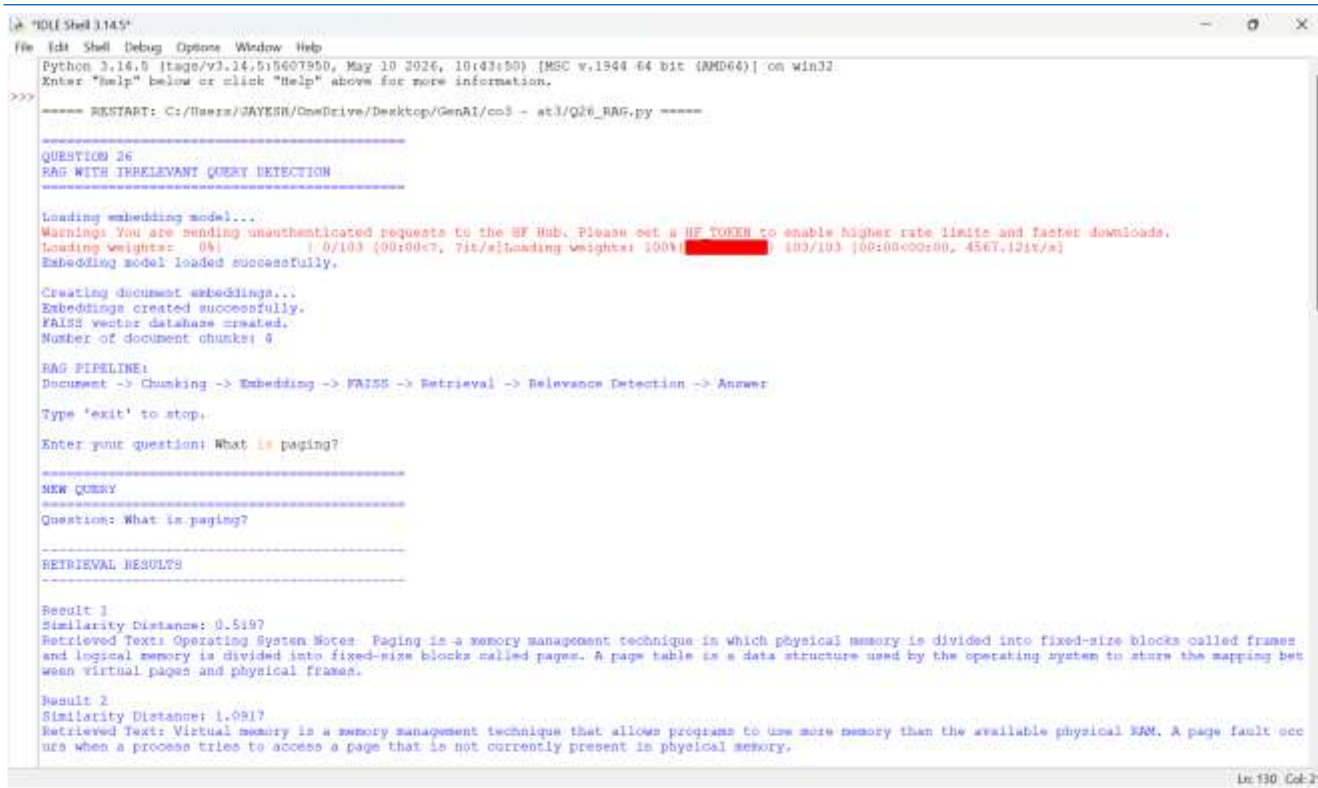
What is virtual memory?

Irrelevant query:

What is the capital of France?

Enter exit to terminate the program.

OUTPUT



```
Question: What is paging?

--------------------------------------------------
RETRIEVAL RESULTS
--------------------------------------------------

Result 1
Similarity Distance: 0.5197
Retrieved Text: Operating System Notes  Paging is a memory management technique in which physical memory is divided into fixed-size blocks called frames
and logical memory is divided into fixed-size blocks called pages. A page table is a data structure used by the operating system to store the mapping bet
ween virtual pages and physical frames.

Result 2
Similarity Distance: 1.0917
Retrieved Text: Virtual memory is a memory management technique that allows programs to use more memory than the available physical RAM. A page fault occ
urs when a process tries to access a page that is not currently present in physical memory.

Result 3
Similarity Distance: 1.2331
Retrieved Text: FIFO page replacement stands for First-In-First-Out page replacement. When a page must be removed from memory, FIFO removes the page that
has been in memory for the longest time.

--------------------------------------------------
RELEVANCE CHECK
--------------------------------------------------
STATUS: RELEVANT QUERY

--------------------------------------------------
FINAL ANSWER
--------------------------------------------------
Operating System Notes  Paging is a memory management technique in which physical memory is divided into fixed-size blocks called frames and logical memo
ry is divided into fixed-size blocks called pages.

Source: Operating System Notes

Enter your question: What is FIFO page replacement?

==================================================
NEW QUERY
==================================================
Question: What is FIFO page replacement?

--------------------------------------------------
RETRIEVAL RESULTS
--------------------------------------------------

Result 1
```

```
--------------------------------------------------
STATUS: RELEVANT QUERY

--------------------------------------------------
FINAL ANSWER
--------------------------------------------------
FIFO page replacement stands for First-In-First-Out page replacement.

Source: Operating System Notes

Enter your question: What is the capital of France?

==================================================
NEW QUERY
==================================================
Question: What is the capital of France?

--------------------------------------------------
RETRIEVAL RESULTS
--------------------------------------------------

Result 1
Similarity Distance: 1.8708
Retrieved Text: FIFO page replacement stands for First-In-First-Out page replacement. When a page must be removed from memory, FIFO removes the page that
has been in memory for the longest time.

Result 2
Similarity Distance: 1.874
Retrieved Text: A process is a program in execution. A thread is a lightweight unit of execution within a process.

Result 3
Similarity Distance: 1.9039
Retrieved Text: Virtual memory is a memory management technique that allows programs to use more memory than the available physical RAM. A page fault occ
urs when a process tries to access a page that is not currently present in physical memory.

--------------------------------------------------
RELEVANCE CHECK
--------------------------------------------------
STATUS: IRRELEVANT QUERY

--------------------------------------------------
FINAL ANSWER
--------------------------------------------------
Information not available in the provided document.
The system did not generate an unsupported answer.

Enter your question:
```

Result:

The program is executed Successfully.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Database Implementation	30%	Correctly builds and queries both FAISS and ChromaDB with accurate understanding of indexing	Builds both with minor issues	Only one database is built and queried correctly	Neither database functions correctly
Retrieval Relevance & Ranking	25%	Retrieval results are relevant and correctly ranked for all test queries	Mostly relevant results across queries	Inconsistent relevance across queries	Results are largely irrelevant
Comparative Analysis	20%	Thoughtful comparison of speed/accuracy trade-offs between the two databases	Reasonable comparison of the two databases	Superficial comparison with little insight	No meaningful comparison attempted
Benchmark Presentation	15%	Clear benchmark results presented (e.g. table or chart)	Results presented adequately	Results presented unclearly	No clear results presented
Methodology Documentation	10%	Clearly documents methodology and test queries used	Adequate documentation of methodology	Minimal documentation	No documentation of methodology

Marks Evaluation:

Question No.	Database Implementation (30%)	Retrieval Relevance & Ranking (25%)	Comparative Analysis (20%)	Benchmark Presentation (15%)	Methodology Documentation (10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	